

Deepak Mallampati

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US-based (willing to relocate to NYC), F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) who builds and deploys agentic LLM systems in production, now looking to bring that hands-on expertise to Inkeep's AI Agent platform -- helping developer-tool customers like Anthropic, PostHog, and Postman succeed with AI agents in their products. Strong TypeScript and Python proficiency, proven ability to enable non-technical users with AI tools, and documented track record of creating technical onboarding materials and knowledge systems.

CORE COMPETENCIES

AI Agent Development & Deployment • TypeScript & Python • RAG System Design • LangChain & LlamaIndex • Customer Enablement • Technical Documentation • FastAPI / Docker • Prompt Engineering • B2B SaaS Context • React / Full-Stack

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Healthcare Technology, AI Consulting | USA

- Built and deployed **agentic LLM systems** for multi-step healthcare queries; improved agent response stability **~25%** and reduced hallucinations **>30%** - same class of AI agents Inkeep's customers need to deploy reliably.
- Designed **RAG systems** for clinical knowledge retrieval; improved contextual retrieval precision **~35%**, reduced irrelevant retrieval **>30%**, and achieved **>90%** retrieval-grounded response alignment.
- Built **AI tools for non-technical clinical staff** (Flask API + web interface for document Q&A); enabled non-engineering teams to adopt AI-assisted search with **>85%** contextual accuracy.
- Created technical documentation, system-limitation guides, and evaluation audit trails for stakeholders.
- Packaged ML/LLM inference as FastAPI/Flask RESTful services with Docker; reduced post-deployment defects **~30%**.

Kent State University - Student Manager

Aug 2023 - May 2025

Operations & Coordination | Ohio, USA

- Led coordination for **150+ students**; structured scheduling and communication reduced coordination delays **~30%**.
- Formalized onboarding for new team members, cutting ramp-up time **~20%** -- directly applicable to customer pilot onboarding.
- Built attendance, participation, and feedback tracking systems surfacing engagement trends.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Applied NLP, Video Analytics | Hyderabad, India

- Built transformer-based video summarization system; reduced manual review time **60-70%**, achieved **~85%** highlight alignment.
- Deployed document Q&A with RAG for non-technical staff; raised contextual accuracy **>85%**.

SELECTED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline with schema-validated JSON contracts, RAG-grounded validation, and audit-ready reasoning traces. Demonstrates production AI agent deployment expertise.

Python, LangChain, LlamaIndex, FastAPI, Docker, RAG

Manga Lens - Chrome Extension for Real-Time AI Translation shipped, Chrome Web Store

Full-stack browser extension with 4 AI vision providers, 7-day translation cache, per-domain configs for 29 sites. Built and shipped independently using TypeScript (Manifest V3, content scripts, service workers).

TypeScript, Manifest V3, Claude Sonnet, GPT-4o mini, multimodal AI

Dream Decoder - Multimodal AI Platform

FastAPI + React/TypeScript/Vite app coordinating multiple AI APIs with intermediate prompt transformation layers; improved contextual alignment ~30%.

FastAPI, React, TypeScript, Vite, Tailwind

EDUCATION

Master's Degree - *Kent State University, Ohio, USA*

Aug 2023 - May 2025

SKILLS

Languages & Frameworks: TypeScript, Python, FastAPI, Flask, React, SQL

LLM & GenAI: LangChain, LlamaIndex, RAG, agentic workflows, prompt engineering, evaluation pipelines, guardrails

ML: scikit-learn, XGBoost, PyTorch, HuggingFace Transformers, embeddings, vector search

Infra: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, observability/logging

Domains: AI Agent Platforms, Developer Tools, Healthcare AI, Customer Enablement